

Quiz # 3

Math 118

Instructor: _____

Fall 2025

Names: _____

To receive full credit you must show all work and provide clear, complete explanations. Use of books, notes, calculators, etc. is not permitted.

1. (5 points) Find the center and radius of the circle:

$$x^2 + 4x + y^2 - 6y + 5 = 0$$

2. (5 points)

(a) In slope intercept form, write the equation of the line that passes through the points $(-2, 2)$ and $(7, 20)$.

(b) Write the equation of the line that is perpendicular to AND has the same y -intercept as the line that you computed in part (a).

3. (5 points) Find the domain of the function:

$$f(x) = \frac{x}{\sqrt{x^2 - 4x - 21}}$$

4. (5 points) Recall that the difference quotient of a function $f(x)$ is:

$$\frac{f(x+h) - f(x)}{h}$$

Find the difference quotient of $f(x) = \frac{1}{x}$.